

**Your Name:**

**ID #:**

**Worksheet: Inverse Trig Functions and their Domains and Ranges**

1. Find the inverse of each function and list the domain and range of  $f^{-1}(x)$

(a)  $f(x) = 5 \sin(x) - 3$  for  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

(b)  $f(\theta) = \frac{1}{2} \tan(\theta)$  for  $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$

(c)  $f(x) = 4 - \cos(2x)$  for  $0 \leq x \leq \frac{\pi}{2}$

(d)  $f(\theta) = 4 - \sin(\theta - \pi) + 1$  for  $\frac{\pi}{2} \leq \theta \leq \frac{3\pi}{2}$

2. Write each trig expression as an algebraic expression and find its domain.

(a)  $\sin \tan^{-1}(x)$

(b)  $\csc \cot^{-1}(x)$

(c)  $\sec \cot^{-1}(x)$

(d)  $\sec \cos^{-1}(x)$